

## Unit – 2 : Session – 7 : SLO – 2 | Single Inheritance

### Activity (Solution)

1. `class Vehicle { String brand = "Honda"; void honk(){ System.out.println("Beep!"); } }`
2. `class Bike extends Vehicle { int gears = 5; void show(){ System.out.println(brand + " bike with " + gears + " gears"); } }`
3. `public class Main { public static void main(String[] a){ Bike b = new Bike(); b.honk(); b.show(); } }`
4. Output: Beep! / Honda bike with 5 gears
5. Why single inheritance: Bike extends exactly ONE superclass (Vehicle). There is a single parent-child link, so it is classified as single inheritance — unlike multiple inheritance (one class, many parents), which Java does not support with classes.